14 FRIEND FUNCTIONS AND CLASSES • THEORY

02 bridging two classes and friend class

Detailed Theory, Exact C++ Rules, Memory Models & Exam Reference

02 — Bridging Two Classes and `friend class`

1. Forward Declaration When Bridging Two Classes

To write a single `friend` function `add_incomes(const teacher&, const researcher&)` that accesses `private` members of two different classes, forward-declare the second class (`class researcher;`) before defining the first class (`class teacher`).

2. `friend class`

Declaring `friend class B;` inside `class A` grants **every member function of** `B` access to `A`'s `private` and `protected` members.